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MILITARY SPECIFICATION

ENAMEL; LUSTRELESS, QUICK-DRYING (FOR AMMUNITION)

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.—This specification covers one type and one grade of a quick-drying synthetic enamel for use as a finishing coat on ammunition.

2. APPLICABLE SPECIFICATIONS, AND STANDARDS

2.1 The following specifications and standards of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT-C-595 —Colors; (For) Ready-Mixed Paints.

TT-P-141 —Paint, Varnish, Lacquer, and Related Materials; General of Inspection, Sampling, and Testing.

TT-P-143 —Paint, Varnish, Lacquer, and Related Materials; General Specification for Packaging, Packing, and Marking.

TT-T-306 —Thinner; Synthetic-Enamel.

TT-T-425 —Titanium Dioxide; Dry (Paint Pigment).

MILITARY

JAN-P-127 —Packaging and Packing for Overseas Shipment; Tape, Adhesive, Pressure-Sensitive, Water Resistant.

JAN-C-490 —Cleaning and Preparation of Ferrous Metal Surfaces for Organic Coatings.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification (type or brand approval).—Unless otherwise specified, enamel purchased under this specification shall be of a brand or type which has been tested and found to meet all requirements of this specification (see 6.4).

3.2 Color.—The enamel shall be furnished in the color as specified in the contract or purchase order, and of a shade conforming to Specification TT-C-595.

3.3 Composition.

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3.3.1 *Vehicle.*—The vehicle shall be a styrenated phthalic alkyd resin (see 6.3), together with the necessary amounts of driers and volatile solvents. The resin solution shall have a color value at 50 percent solids content no darker than 10 (Gardner Color Standards of 1933). Small amounts of antioxidants, wetting agents, and stabilizers may be present. The vehicle shall show a negative rosin and phenolic resin test.

TABLE I.—*Pigmentation.*

Color, number (TT–C–595)	Color	Pigmentation[1]
3115	Red	Toluidine Red, titanium dioxide[2].
[3]32205	Orange	Chrome orange, molybdate orange.
3305	Yellow	C.P. medium chrome yellow, yellow iron oxide.
[3]32405	Green	Chrome green.
3412	Olive drab	Red or yellow iron oxide, carbon or lampblack, C.P. medium chrome yellow, Titanium dioxide.[2]
[3]32515	Medium blue	Milori blue, titanium dioxide[2], lampblack, yellow iron oxide.
3530	Light blue	Milori blue, titanium dioxide[2], C.P. chrome yellow.
3610	Ocean gray	Titanium dioxide[2], carbon or lampblack.
3615	Army: Blue gray Navy: Slate	Titanium dioxide[2], carbon or lampblack, yellow iron oxide.
[4]3715	White	Titanium dioxide[2].
3725	Black	Black iron oxide, carbon or lampblack.
3730	Purple	Titanium dioxide[2], thioindigoid maroon. Dichloro - iso - dibenzanthrone violet.

[1] Exclusive of extender pigments. Small amounts of thio-indigo red, phthalocyanine blue, indanthrene blue, and inorganic tinting pigments may be used to match the color chip, provided these pigments are of high durability.

[2] Rutile chalk resisting titanium dioxide conforming to Specification TT–T–425, type III, class C.

[3] This color is listed in Specification TT–C–595 as a semigloss color.

[4] The enamel shall meet the reflectance requirement in table II.

3.3.2 *Pigments.*—Hiding pigments shall be "pure" (free from extenders), that is, the chrome pigments shall be the "C.P." type, the iron oxides (red and yellow) shall be practically pure iron oxide, etc. The permissible hiding pigments for each color shall be as shown in Table I. Extender pigments shall be siliceous materials or siliceous materials and barytes and shall not exceed the amounts specified in Table II. The amount of barytes used shall not exceed 20 percent of the extender pigments by weight. Calcium sulfate or carbonates shall not be employed alone or as a component of any pigment.

3.4 Quantitative requirements.—The enamel shall meet the quantitative requirements specified in Table II and in 3.4.1.

3.4.1 *Dry opacity.*—The enamel shall show a minimum dry-film contrast ratio as specified in Table III, when applied at the wet film thickness indicated.

3.5 Qualitative requirements.

3.5.1 *Condition in container.*—The enamel show no excessive settling in a freshly opened full can, and shall mix readily to a smooth homogeneous state. The enamel shall show no curdling, livering, hard caking or tough gummy sediment, and shall be free from lumps and skins.

3.5.2 *Storage stability.*

3.5.2.1 *Partially-full container.*—After 48 hours standing at a temperature between 21° and 32° C. (70° and 90° F.), a three-quarter-filled, closed 8-ounce glass jar of the material shall show no skinning. After 30 days standing, the material shall show no livering, curdling, hard caking or tough gummy sediment. The enamel shall mix readily to a smooth homogeneous state, and any skin formed shall be continuous and easily removed.

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TABLE II.—*Quantitative requirements.*

Characteristics	Requirements	
	Minimum	Maximum
Total solids, percent by weight of enamel—		
Orange No. 2205, Yellow No. 3305, Olive Drab No. 3412, Light Blue No. 3530, Ocean Gray No. 3610, Blue Gray (Slate) No. 3615, White No. 3715, Purple No. 3730	55	
Red No. 3115, Green No. 2405, Medium Blue No. 2515, Black No. 3725	52	
Pigment, percent by weight of enamel—		
Orange No. 2205, Yellow No. 3305, Light Blue No. 3530, Blue Gray (Slate) No. 3615, White No. 3715	35	39
Green No. 2405, Olive Drab No. 3412, Ocean Gray No. 3610, Black No. 3725, Purple No. 3730	33	37
Red No. 3115, Medium Blue No. 2515	29	33
Vehicle solids, percent by weight of enamel—		
Red No. 3115	21	24
All other colors	17	21
Pigment volume, percent of total solids volume—		
Red No. 3115, Green No. 2405, Olive Drab No. 3412, Medium Blue No. 2515, Ocean Gray No. 3610, Black No. 3725, Purple No. 3730		39
Orange No. 2205, Yellow No. 3305, Light Blue No. 3530, Blue Gray (Slate) No. 3615, White No. 3715		42
Phthalic anhydride, percent by weight of nonvolatile vehicle	18	26
Extender pigment, percent by weight of total pigment—		
White No. 3715		50
Orange No. 2205, Yellow No. 3305, Blue Gray (Slate) No. 3615		55
Green No. 2405, Olive Drab No. 3412, Medium Blue No. 2515, Light Blue No. 3530, Ocean Gray No. 3610, Purple No. 3730		65
Red No. 3115		70
Black No. 3725		83
Iron Oxide (Fe_2O_3—by analysis), percent by weight of pigment—		
Olive Drab No. 3412	30[1]	
Iron Oxide (Fe_3O_4—by analysis), percent by weight of pigment—		
Black No. 3725	15	
Titanium dioxide (TiO_2—by analysis) percent by weight of pigment—		
White No. 3715	47	
Blue Gray (Slate) No. 3615	37	

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TABLE II.—*Quantitative requirements.*—Continued

Characteristics	Requirements	
	Minimum	Maximum
Ocean Gray No. 3610	30	
Medium Blue No. 2515, Light Blue No. 3530, Purple No. 3730	24	
Lead Chromate ($PbCrO_4$—by analysis), percent by weight of pigment—		
Yellow No. 3305	37	
Water, percent by weight of enamel		1.0
Coarse particles and skins (retained on No. 325 mesh sieve), percent by weight of pigment		0.5
Apparent reflectivity (White enamel), percent	80	
60° specular gloss	2	8
Consistency (package), Krebs-Stormer shearing rate—		
200 rpm		
Grams	125	175
Equivalent KU	67	77
Consistency (reduced 2 parts by volume of thinner conforming to Specification TT–T–306 to 5 parts by volume package material), No. 4 Ford Cup, seconds	15	25
Fineness of Grind	5	
Drying time, air drying—		
Set to touch, minutes	3	6
Dry hard, minutes		10
After-tack free, minutes		15
Dry through, minutes		20
Full hardness, hours		72

[1] Lead chromate ($PbCrO_4$) may be substituted on an equal weight basis.

3.5.2.2 *Full container.* — After 6 months standing under warehouse storage conditions, a full, quart container of the enamel shall show no skinning, livering, curdling, hard caking or tough gummy sediment. The enamel shall remix readily to a smooth homogeneous state and shall have a maximum consistency of 89 KU. The enamel shall meet all other requirements of the specification.

3.5.3 *Dilution stability.*—The material shall show no evidence of precipitation and shall remain stable and uniform after reduction of one part by volume of the package material with one part by volume of thinner conforming to Specification TT–T–306.

3.5.4 *Suspension properties.* — Enamel thinned for spraying shall show no more than slight settling, no caking, and shall redisperse to a smooth homogeneous state when tested as in 4.4.3.

3.5.5 *Brushing properties.* — The enamel shall be of brushing consistency in the package condition and shall be capable of being brushed out and laid off quickly on a 4 by 12 inch steel panel using a 1½-inch brush.

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TABLE III.—*Dry opacity.*

Color	Wet film thickness, mils, maximum	Contrast ratio of dry film, minimum
Red No. 3115	3.0[1]	0.91
Orange No. 2205	3.0	0.86
Yellow No. 3305	3.0	0.96
Green No. 2405	2.0[2]	0.98
Olive Drab No. 3412	2.0	0.99
Medium Blue No. 2515	2.0	0.98
Light Blue No. 3530	2.0	0.98
Ocean Gray No. 3610	2.0	0.99
Blue Gray (Slate) No. 3615.	2.0	0.98
White No. 3715	3.0	0.95
Black No. 3725	2.0	0.99
Purple No. 3730	2.0	0.98

[1] Apply enamel as received by drawdown using a 0.003-inch (0.006-inch gap clearance) film applicator.
[2] Apply enamel as received by drawdown using a 0.002-inch (0.004-inch gap clearance) film applicator.

3.5.6 *Dipping properties.* — The enamel when reduced with 1.0 ± 0.2 parts by volume of thinner conforming to Specification TT–T–306 to 3 parts by volume of package material shall show satisfactory dipping properties. A 4 by 12 inch steel panel dipped its full length in the thinned enamel shall present a uniform appearance free from color separation, sagging, running, or streaking.

3.5.7 *Spraying properties; appearance when dry.*—The enamel when reduced with 2 parts by volume of thinner conforming to Specification TT–T–306 to 5 parts by volume of package material shall spray satisfactorily in all respects, and shall show no dusting, mottling or color separation and shall present a smooth, lustreless finish free from grit, seediness, or other surface imperfections.

3.5.8 *Flexibility.*—A film of the enamel prepared and tested as in 4.4.5 shall withstand bending without cracking or flaking.

3.5.9 *Knife test.*—The enamel shall adhere tightly and not flake, crack, or powder from the metal when tested as in 4.4.6. The cut shall show beveled edges.

3.5.10 *Adhesion.*—After air drying for one hour a film of the enamel prepared and tested as in 4.4.7 shall show no removal of the coating by the adhesive tape beyond one-sixteenth inch on either side of the score line.

3.5.11 *Water resistance.* — A film of the enamel prepared and tested as in 4.4.8 shall show no wrinkling or blistering immediately after removal of the panel from the water. The enamel shall be no more than slightly affected when examined 2 hours after removal. After 24 hour air drying, the portion of the panel which was immersed shall be almost indistinguishable from the portion which was not immersed with regard to hardness, color, and gloss.

3.5.12 *Recoating.*—A film of enamel prepared and tested as in 4.4.9 shall not blister, wrinkle, or show other evidence of lifting when recoated with white enamel. With the exception of Red No. 3115, the film shall contain no bleeding pigments which will discolor the white enamel.

3.5.13 *Salt spray resistance.*—A film of enamel prepared and tested as in 4.4.10 and examined immediately after removal from the test shall show no rust creepage or undercutting beyond one-eighth inch from the scratch mark. At all other points of the panel there shall be no more than a trace of rusting (see 6.5) and no more than 5 scattered blisters no larger than 1 mm in diameter. After removal of the enamel, the surface of the steel shall show no more than a trace of rusting, pitting, or corrosion.

3.5.14 *Accelerated Weathering.*—A film of the enamel prepared and tested as in 4.4.11 shall show no more than slight chalking (see 6.5) and a color change equivalent to a lightness-difference estimate not exceeding 4 units.

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3.5.15 *Weather resistance.* — A film of enamel prepared and exposed as in 4.4.12 shall show no checking, cracking, or appreciable film deterioration. There shall be no more than moderate chalking (see 6.5) of colors Light Blue No. 3530, Gray No. 3615 and White No. 3715 and no more than light chalking of all other colors. There shall be no rust creepage or blistering beyond one-eighth inch from the score mark and no more than a trace of rusting (see 6.5) at all other points. The film shall show no excessive change in value or chroma and no change in hue. After removal of any chalking which has occurred, the original color shall be substantially restored and the washed area shall show no more than slight fading or darkening.

3.5.16 *Toxicity.*—The product shall contain no benzol (benzene), methanol, chlorinated solvents, or other compounds of a highly toxic nature.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures.—For naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material. For Air Force purchases, inspection shall be in accordance with Section 1000 of Specification TT–P–141. For all other purchases unless otherwise specified, inspection shall be performed in accordance with method 101.1 of Specification TT–P–141.

4.2 Sampling

4.2.1 *Lot.*—For purposes of sampling, a lot shall consist of enamel from one batch offered for delivery at one time.

4.2.2 *Sampling for tests (at a Government laboratory).*—From each inspection lot (see 4.2.1) the Government inspector shall select two containers at random. From each of the two containers, 1 quart specimens shall be taken and placed in separate, clean, dry, metal or glass containers, and then sealed, marked and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.2.2.1 The Government inspector shall, when requested by the contractor, select sample specimens in accordance with 4.2.2. with the following exception: Three quart specimens shall be taken instead of 1 quart specimens. Each three quart specimen shall be divided into three equal parts, one part to be delivered to the contractor, one part to the testing laboratory, and one part to be held by the inspector in case of dispute.

4.2.2.2 *Ingredient materials.*—One pint of the styrenated alkyd resin, and when and as requested by the laboratory officer or other authority, samples of other ingredient materials shall be taken by the Government inspector for test purposes.

4.2.3 *Testing activity.*—Unless otherwise specified or designated (see 4.2.2), the samples taken by the Government inspector as required for lot acceptance test purposes, shall be forwarded to the Laboratory Officer: Industrial Test Laboratory, Building 121, Philadelphia Naval Shipyard, Philadelphia 12, Pa.; Industrial Laboratory, Mare Island Naval Shipyard, Vallejo, Calif.; Chemical Laboratory, Norfolk Naval Shipyard, Portsmouth, Va.; or to the Material Laboratory, New York Naval Shipyard, Brooklyn, N. Y.

4.2.4 *Sampling for inspection of containers (at place of manufacture).*—A random sample of filled containers shall be selected by the Government inspector in accordance with Standard MIL–STD–105 at inspection level I and acceptable quality level = 2.5 percent defective to verify compliance with this specification in regard to fill, closure, marking, and other requirements not involving tests.

4.3 Inspection and tests.

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4.3.1 *Inspection of containers (at place of manufacture).*—Each sample filled container as specified in 4.2.4 shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the the amount of contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests, provided the contractor has removed (or reworked) all nonconforming products.

4.3.2 *Testing.*—Testing under this specification shall be for the following purposes:

(a) Qualification.

(b) Acceptance of individual lots.

Qualification testing shall consist of all tests covered by this specification (see 6.4). Acceptance testing shall normally consist of all tests with the exception of 3.5.2.2 and 3.5.15. The right is reserved to make any additional tests deemed necessary to determine that the enamel meets the requirements of the specification.

4.3.3 *Lot acceptance tests.*—The samples selected in accordance with 4.2.2 shall be subjected to the tests specified in 4.3.2 for acceptance testing, except that the laboratory may perform complete tests on only one sample provided sufficient tests are made on the second to verify that both are alike. Lots shall be accepted or rejected by the Government inspector on the basis of the laboratory test report on the transmitted samples. Rejected lots may be resubmitted for acceptance tests, provided that the contractor has removed (or reworked) all nonconforming products.

4.4 Test procedures.

4.4.1 The following tests shall be conducted in accordance with Specification TT–P–141:

Tests	*Method*
Color of transparent liquids	424.8
Color of pigmented coatings	[1]423.1
Isolation of vehicle (supercentrifuge)	403.2
Rosin in isolated vehicle	503.1
Phenolic resin	514.1
Total solids	404.1
Pigment content	402.2
Vehicle solids	404.1
Pigment volume	431.2
Phthalic anhydride	[2]702.1
Water in paints and varnishes	408.2
Coarse particles and skins	409.2
Apparent reflectivity	612.1
Specular gloss	[3]610.1
Consistency	
Krebs-Stormer	428.1
Ford Cup	428.2
Fineness of grind	441.1
Dry opacity	412.1
Condition in container	301.1
Skinning	414.1
Storage Stability	414.2
Dilution stability	439.1
Brushing properties	432.1
Dipping properties	434.1
Spraying properties	433.1
Benzol	509.1

[1] Spray the enamel to complete hiding and allow to dry overnight.

[2] Procedure A modified as follows: Add 50 milliliters of benzol instead of 10 milliliters and saponify with 60 milliliters of 1.0 N alcoholic potassium hydroxide. Before filtering add an additional 50 milliliters of benzol and warm sample to about 50° C. Filter while hot using equal volumes of benzol and absolute alcohol as wash medium. Do not use ether.

[3] Apply the enamel as received by drawdown using a 0.0015-inch (0.003-inch gap clearance) film applicator.

4.4.2 *Analysis of pigment.* — Extract the pigment as in method 402.1 of Specification TT–P–141 and analyze as below. Determine the absence of carbonates and calcium sulfate in the extender pigment by appropriate qualitative tests.

4.4.2.1 *Olive drab No. 3412.*—Determine iron oxide, Fe_2O_3, content of the extracted pigment in accordance with method 714.1, section 1. Determine lead chromate, $PbCrO_4$, in accordance with method 713.1, section 4,

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using double the specified quantity of sample, potassium hydroxide and hydrochloric acid.

4.4.2.2 *Black No. 3725.*—Determine iron oxide, Fe_3O_4, content of the extracted pigment in accordance with method 714.1, section 1, calculating total iron to Fe_3O_4.

4.4.2.3 *White No. 3715, blue gray (slate) No. 3615, ocean gray No. 3610, medium blue No. 2515, light blue No. 3530, purple No. 3730.*—Determine titanium dioxide (TiO_2) content of the extracted pigment in accordance with method 708.2, section 1.

4.4.2.4 *Yellow No. 3305.*—Determine lead chromate, $PbCrO_4$, content of the extracted pigment in accordance with method 713.1, sections 2 and 4.

4.4.2.5 *Other colors.*—Appropriate qualitative and quantitative tests shall be made on the extracted pigments to determine if permissible pigments were used in formulating the other enamels.

4.4.3 *Suspension properties.*—The enamel shall be thinned to spraying consistency as specified in 3.5.7. Six ounces of the reduced material shall be placed in an 8-ounce glass jar. The stoppered jar shall not be agitated or disturbed for 24 hours. At the end of this period examine the material for hard or excessive settling by means of a spatula. Do not stir. Restopper the jar and shake vigorously for no longer than 20 seconds. Re-examine the material for any evidence of nonhomogeneity or undispersed pigment.

4.4.4 *Drying time.*

4.4.4.1 Determine set to touch, dry hard and after-tack-free in accordance with method 406.1, section 1, of Specification TT–P–141. As a referee test for dry hard, the enamel applied as in method 406.1 shall be print-free at the end of 10 minutes using a uniform pressure of one pound per square inch applied for 20 minutes. The print test shall be performed at 25° ± 1° C. in accordance with method 621.1 except that no felt shall be interposed between the weight and the cheese cloth. Disregard any marks which do not remain noticeable after the film is lightly polished with a soft cloth.

4.4.4.2 Determine dry through and full hardness in accordance with method 406.1, section 2, except that the enamel shall be drawn down with a 0.002-inch (0.004-inch gap clearance) film applicator upon a steel panel which has been phosphate cleaned and prepared in accordance with method 201.1.

4.4.5 *Flexibility.*—Determine flexibility in accordance with method 622.1 of Specification TT–P–141. Apply the enamel to a flat tinplate panel of approximately 31 gage by means of a 0.002-inch (0.004-inch gap clearance) film applicator. Bake the panel for 24 hours at 105° ± 2° C. (221° ± 4° F.). Condition the panel for one-half hour at 25° ± 1° C. Bend over a ¼-inch mandrel and examine for defects as described in 3.5.8 using a 7-diameter magnification.

4.4.6 *Knife test.*—Perform the knife test in accordance with method 617.1 of Specification TT–P–141 using a flat portion of the panel from the flexibility test. Examine the panel for defects described in 3.5.9.

4.4.7 *Adhesion.*

4.4.7.1 Using a 0.0020-inch (0.0040-inch gap clearance) film applicator, draw down a 2-inch wide film of the enamel on a steel panel which has been phosphate cleaned in accordance with method 201.1 of Specification TT–P–141.

4.4.7.2 Air dry the specimen for one hour at 25° ± 1° C. and then score a line through to the metal across the width of the film using a sharp pointed knife. The film shall then be taped perpendicular to and across the score line with water-resistant, pressure-sensitive adhesive tape (¾-inch width) conforming to the requirements of Specifica-

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tion JAN–P–127, type II or III, grade A or B. The tape shall be pressed in firm contact with the film and shall extend for approximately one inch on each side of the score line. All air bubbles shall be rolled out by firm pressure of the thumb. Allow approximately 10 seconds for the test area to return to room temperature. Grasp a free end of the tape and at a rapid speed strip it from the specimen by pulling the tape back upon itself at 180°. Observe the specimen for compliance with the requirements of 3.5.10.

4.4.8 *Water resistance.*—Prepare a film of the enamel as in 4.4.7.1 and air dry for 72 hours. All exposed, uncoated metal surfaces shall be coated with wax or other suitable coating and the panel shall be immersed for 18 hours in distilled water at 25° ± 1° C. (77° ± 2° F.) in accordance with method 601.1 of Specification TT–P–141. At the end of the test period remove the panel from the water and inspect for compliance with the requirements of 3.5.11.

4.4.9 *Recoating.*—Prepare two test panels as in 4.4.7.1 and air dry for 24 and 96 hours respectively. At the end of its drying period immerse each panel to a depth of approximately 2½ inches in white enamel conforming to this specification which has been reduced with one part by volume thinner conforming to Specification TT–T–306 to four parts by volume of package material. At the end of 5 seconds remove the panel, dry in a vertical position and examine for compliance with 3.5.12.

4.4.10 *Salt spray resistance.*—Three 4- x 12-inch steel panels, solvent cleaned in accordance with method 201.1, shall be given a phosphate coating conforming to the requirements of Specification JAN–C–490, grade I. Spray a coat of the enamel to a uniform dry film thickness of 0.0009 to 0.0011 inch. Air dry for 96 hours, score all specimens and expose to the salt spray for 144 hours as described in method 606.1. Upon removal, wash the panels gently in running water not warmer than 100° F. until free from any visible salt deposits and examine immediately for compliance with the requirements of 3.5.13. Strip the enamel film from the panels by means of lacquer thinner and inspect the steel for rust, pitting, or corrosion.

4.4.11 *Accelerated weathering.*—Prepare a film of the enamel as in 4.4.5 and air dry for 72 hours. Expose the panel for 168 hours to accelerated weathering in accordance with method 615.2 of Specification TT–P–141. Examine the exposed panel for chalking by rubbing with a piece of velvet or cheesecloth wrapped around the finger. The rubbing shall be done under moderate pressure by drawing the cloth across the width of the panel in two different locations. Examine the panel for color change by measuring the apparent reflectivity (method 612.1) before and after exposure, and calculate the lightness-difference estimate ΔL from the expression:

$$\Delta L = K \Delta (y^{1/2})$$

where:

y = the luminous apparent reflectance of the panel measured before and after the test; and

K = 100 (K = 10 if reflectance values are expressed in percent).

The method for determining the lightness difference estimate is described in detail in example 4, National Bureau of Standards Circular C429, Photoelectric Tristimulus Colorimetry with Three Filters.

4.4.12 *Weather resistance.*—Prepare two scored test panels as in 4.4.10. Air dry for 96 hours and place on outdoor exposure for 18 months at an angle of 45° South in the latitude of Washington, D. C. At the conclusion of the exposure period, inspect the panels for compliance with 3.5.15. Determine chalking as in 4.4.11. Wash the panels with a warm soap solution using a soft sponge or cloth, rinse, dry and examine for color change.

4.4.13 *Toxicity.*—Note the odor and other characteristics of the enamel during its ex-

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amination and if there is reason to suspect the presence of any of the prohibited solvents, make appropriate confirmatory tests.

5. PREPARATION FOR DELIVERY

5.1 The enamel shall be delivered in containers of the size specified in the contract or order (see 6.2). Unless otherwise specified, preparation for delivery shall conform to Specification TT–P–143, except that marking for all Military Services shall be applied in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended Uses.**—The enamel covered by this specification is intended for use primarily as a finishing coat for shell, bombs, grenades, and other ammunition components. *IT IS NOT INTENDED* for use where gasoline resistance is a prime requisite such as automotive equipment.

6.2 **Ordering data.**—Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of container required and whether preparation for domestic or oversea shipment is required (see 5.1).
(c) Color

6.3 The olive drab enamel is contemplated to be comparable in performance to the following approximate composition by weight:

No. 3412 Olive drab Enamel
347 lbs. Synthetic yellow orange iron oxide
18 lbs. Lampblack
190 lbs. Fibrous magnesium silicate
445 lbs. Acicular talc
335 lbs. Styrenated alkyd resin (Z1–Z4 at 50 per cent solids in MS)
400 lbs. Xylol
3 lbs. Diethylamine

Grind 18 hours in a steel ball mill using 4–1 ratio of steel balls by weight and reduce as follows:

685 lbs. Styrenated alkyd resin (Z1–Z3 at 60 per cent solids in xylol)
445 lbs. Xylol
3 lbs. Cobalt naphthenate
3 lbs. Anti skinning agent

At the lower viscosities, the use of small amounts of suspension agents will improve settling properties.

6.3.1 The styrenated alkyd resins used in these formulations are copolymers of styrene and phthalic alkyd resins modified with drying vegetable oils. It has generally been found that the finished enamel must be finely ground in order to insure adequate salt spray resistance and good ballistic performance when applied on ammunition.

6.3.2 Enamels of this type show some tendency to be thixotropic at higher viscosities. The use of a material such as diethylamine to the extent of 0.5–1.0 per cent on the resin solids basis has been found to reduce this condition materially and improve the package stability.

6.4 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been tested and found satisfactory for inclusion on the Military Qualified Products List. Information concerning qualification may be obtained by communicating with the Commanding General, Aberdeen Proving Ground, Maryland, Attention: Paint and Chemical Laboratory.

6.5 It is intended that the maximum amount of rusting in the salt spray test (see 3.5.13) and the weather resistance test (see 3.5.15) be comparable to Photo 9–1 in ASTM Standard Method of Evaluating Degree of Resistance to Rusting Obtained with Paint on Iron or Steel Surfaces D 610–43 and that the maximum amount of chalking in the

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accelerated weather test (see 3.5.14) and weather resistance test be comparable to Photo 8 for slight chalking, Photo 6 for light chalking and Photo 4 for moderate chalking in ASTM Standard Method of Evaluating Degree of Resistance to Chalking of Exterior Paint of the Linseed Oil Type D 659–44. Lighter colored cloth facilitates the evaluation of chalk resistance of darker colors. These standards may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.

6.6 Thinners conforming to Specifications TT–T–306 and TT–N–97, type I, grade B may be used interchangeably depending on climatic conditions. TT–T–306 is recommended for use during the warmer months of the year as a safeguard against dry overspray. TT–N–97, type I, grade B is recommended for use during cold winter months or when the humidity is excessively high in order to prevent undue loss of drying properties.

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Custodians:
Army—Ordnance Corps
Navy—Bureau of Ships
Air Force

Other interest:
Army—ESigTC
Navy—SOr.

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